

$N=1$  susy: Chiral Model:  $\psi$ : Weyl,  $f \in \mathbb{R}$

$$\left\{ \begin{array}{l} \mathcal{L}_0 = i\psi^\dagger \bar{\sigma}^\mu \partial_\mu \psi - \partial_\mu f \partial^\mu f \\ \delta_\epsilon \psi = \epsilon \psi \quad \delta_\epsilon \bar{\psi} = \epsilon^\dagger \psi^\dagger \\ \delta_\epsilon \psi_\alpha = -i\sigma^\mu_{\alpha\dot{\alpha}} \epsilon^{\dagger\dot{\alpha}} \partial_\mu f \quad \delta_\epsilon \psi^\dagger_{\dot{\alpha}} = i\partial_\mu f \epsilon^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \end{array} \right\}$$

$$\{Q_\alpha, Q^{\dagger\dot{\alpha}}\} = -2\sigma^\mu_{\alpha\dot{\alpha}} P_\mu \quad \epsilon_M = \begin{pmatrix} \epsilon_\alpha \\ \epsilon^{\dagger\dot{\alpha}} \end{pmatrix}$$

$$\bullet \Psi_M = \begin{pmatrix} \psi_\alpha \\ \psi^{\dagger\dot{\alpha}} \end{pmatrix}$$

$$\phi(x) = \int d\vec{p} [a_-(p) e^{ipx} + a_+^\dagger e^{-ipx}]$$

$$\psi_\alpha(x) = P_L \Psi_M(x) = \sum_{s=\pm} \int d\vec{p} (b_s(p) P_L u_s(p) e^{ipx} + b_s^\dagger(p) P_L v_s(p) e^{-ipx})$$

with

$$(\psi^\dagger)^{\dot{\alpha}} = P_R \Psi_M$$

$$[a_\pm(p), a_\pm^\dagger(p')] = (2\pi)^3 2E_p \delta^3(\vec{p} - \vec{p}')$$

$$\{b_\pm(p), b_\pm^\dagger(p')\} = (2\pi)^3 2E_p \delta^3(\vec{p} - \vec{p}')$$

Recall:  $v_s(p) = \begin{pmatrix} |p\rangle_\alpha \\ |p\rangle_{\dot{\alpha}} \end{pmatrix} \quad | \quad u_s(p) = \begin{pmatrix} \langle p|^\alpha \\ \langle p|_{\dot{\alpha}} \end{pmatrix}$

$$P_L v_s = |p\rangle \delta_{s+}, \quad P_R v_s = |p\rangle \delta_{s-}$$

Define Anticommuting spinors  $\zeta$   $\langle \epsilon |_\alpha \equiv \epsilon^\dagger_{\dot{\alpha}}$   
 $|\epsilon\rangle_\alpha \equiv \epsilon_\alpha$

Now,  $\delta_\epsilon$  acts on the OPERATOR Algebra, thus  $\delta_\epsilon$  acts on  $a_\pm, b_\pm$ .

$$\begin{aligned} \delta_\epsilon \phi &= \int d\vec{p} (\delta_\epsilon a_- | e^{ipx} + (\delta_\epsilon a_+^\dagger) e^{-ipx}) \\ &= \langle \epsilon |^\alpha \left( \sum_{s=\pm} \int d\vec{p} [b_s P_L u_s e^{ipx} + b_s^\dagger(p) P_L v_s e^{-ipx}] \right) \end{aligned}$$

$$u_\pm = v_\mp \Rightarrow u_+ = \begin{pmatrix} 0 \\ |p\rangle_{\dot{\alpha}} \end{pmatrix}, \quad u_- = \begin{pmatrix} |p\rangle_\alpha \\ 0 \end{pmatrix}$$

$$\cdot \quad \zeta_\alpha = \langle \epsilon |_\alpha |p\rangle_\alpha, \quad b |p\rangle$$

$$\begin{aligned}
 (\delta_\epsilon a_+^\dagger) &= [\epsilon | \alpha | P]_\alpha b_+^\dagger(P) \\
 \rightarrow \text{Take } \dagger \text{ of both} \\
 ([\epsilon P])^\dagger &= \langle \epsilon P \rangle \\
 \therefore \delta_\epsilon a_+(P) &= \langle \epsilon P \rangle b_+(P)
 \end{aligned}$$

$$\delta_\epsilon a_-(P) = [\epsilon P] b_-(P), \quad \delta_\epsilon a_+(P) = \langle \epsilon P \rangle b_+(P)$$

$$\begin{aligned}
 * \text{ Since } \delta_\epsilon \psi_\alpha &= -i \sigma_{\alpha\dot{\alpha}}^\mu (\epsilon^\dagger)^{\dot{\alpha}} \partial_\mu \phi \\
 \delta_\epsilon b_s P_L u_s &= -i \sigma_{\alpha\dot{\alpha}}^\mu \langle \epsilon | \dot{\alpha} | i P_\mu \rangle u_{-}(P) \\
 &= \langle \epsilon | \dot{\alpha} | P_{\alpha\dot{\alpha}} \rangle u_{-}(P) \\
 \delta_\epsilon b_{-}(P) [P]_\alpha &= u_{-}(P) P_{\alpha\dot{\alpha}} [\epsilon]^\alpha \\
 \delta_\epsilon b_{-}(P) [\epsilon P] &= -u_{-}(P) [\epsilon | P]_\alpha \langle P | \dot{\alpha} | \epsilon \rangle^\alpha \\
 \delta_\epsilon b_{-}(P) &= -u_{-}(P) \langle \epsilon P \rangle \\
 &= \langle \epsilon P \rangle u_{-}(P)
 \end{aligned}$$

$$\begin{aligned}
 * \delta_\epsilon b_{-}(P) &= \langle \epsilon P \rangle u_{-}(P) \\
 \delta_\epsilon b_{+}(P) &= [\epsilon P] a_{+}(P)
 \end{aligned}$$

$$* \text{ Claim: } [\delta_{\epsilon_1}, \delta_{\epsilon_2}] \psi(P) = \epsilon_1^\mu P_\mu \psi(P)$$

$\psi(P)$ : Creation/ann. op.

Proof: Will do one  $a$  and one  $b$

$$\begin{aligned}
 [\delta_1, \delta_2] u_{-}(P) &= \delta_1([\epsilon_2 P] b_{-}(P)) - \delta_2([\epsilon_1 P] b_{-}(P)) \\
 &= [\epsilon_1 P] \langle \epsilon_2 P \rangle u_{-} - [\epsilon_2 P] \langle \epsilon_1 P \rangle u_{-} \\
 &= (-[\epsilon_1 | P] \langle \epsilon_2 P |) u_{-} + ([\epsilon_2 | P] \langle \epsilon_1 P |) u_{-} \\
 &= ([\epsilon_1 |^\alpha P_{\alpha\dot{\alpha}} | \epsilon_2 \rangle^{\dot{\alpha}} - [\epsilon_2 |^\alpha P_{\alpha\dot{\alpha}} | \epsilon_1 \rangle^{\dot{\alpha}}) u_{-} \\
 &\quad - \langle \epsilon_1 | P | \epsilon_2 \rangle \\
 &= (\epsilon_1^\alpha \sigma_{\alpha\dot{\alpha}}^\mu \epsilon_2^{\dagger\dot{\alpha}} - \epsilon_2^{\dagger\dot{\alpha}} (\sigma^\mu)^{\dot{\alpha}\alpha} \epsilon_{1\alpha}) P_\mu u_{-}
 \end{aligned}$$

$$\epsilon_{\mu,2} \gamma^\mu \epsilon_{\mu,1}$$

Subproof

$$\begin{aligned}
 & \begin{pmatrix} \epsilon_2^\alpha \\ \epsilon_2^{\dagger\dot{\alpha}} \end{pmatrix} \begin{pmatrix} 0 & \sigma^\mu_{\alpha\dot{\alpha}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} & 0 \end{pmatrix} \begin{pmatrix} \epsilon_1^\beta \\ \epsilon_1^{\dagger\dot{\beta}} \end{pmatrix} \\
 &= \begin{pmatrix} \epsilon_2^\alpha & \epsilon_2^{\dagger\dot{\alpha}} \end{pmatrix} \begin{pmatrix} \sigma^\mu_{\alpha\dot{\alpha}} \epsilon_1^{\dagger\dot{\alpha}} \\ (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \epsilon_{1\alpha} \end{pmatrix} \\
 &= \epsilon_2^\alpha \sigma^\mu_{\alpha\dot{\alpha}} \epsilon_1^{\dagger\dot{\alpha}} + \epsilon_2^{\dagger\dot{\alpha}} (\bar{\sigma}^\mu)^{\dot{\alpha}\alpha} \epsilon_{1\alpha}
 \end{aligned}$$

$$\begin{aligned}
 [2|P|1] &= -[2P]\langle 1| = -\langle 1P\rangle [P2] \\
 &= \langle 1|(-|P\rangle [P1])|2] \\
 &= \langle 1|P|2]
 \end{aligned}$$

and close ... off by a - sign.

$$[\delta_1, \delta_2] \psi(p) = a^\mu p_\mu \psi(p), \quad a^\mu = \epsilon_{\mu 12} \gamma^\mu \epsilon_{\mu 11}$$

\* The supersymmetry generators can be written in terms of creation/ann. ops as we write

$$Q_\mu = \begin{pmatrix} |Q\rangle_\alpha \\ |Q\rangle^{\dot{\alpha}} \end{pmatrix}, \quad E_\mu = \begin{pmatrix} |E\rangle_\alpha \\ |E\rangle^{\dot{\alpha}} \end{pmatrix}$$

$$\begin{aligned}
 \text{and since } \delta_\epsilon \psi &= [\bar{E}_\mu Q_\mu, \psi] \quad | \quad \bar{E} = (|E\rangle_\alpha, |E\rangle^{\dot{\alpha}}) \\
 \delta_\epsilon \psi &= [|E Q\rangle + |E Q\rangle, \psi] \\
 \delta_\epsilon \psi &= (|E Q\rangle + |E Q\rangle) \psi \\
 \delta_\epsilon \phi &= [E \psi] = (|E Q\rangle + |E Q\rangle) \phi
 \end{aligned}$$

\* I don't think this is a proper derivation but:

$$\delta_\epsilon a_+^\dagger = [E p] b_+^\dagger(p) = |E Q\rangle a_+^\dagger - a_+^\dagger |E Q\rangle$$

$$\begin{aligned}
 & \therefore [P]_\alpha b_+^\dagger(p) a_+ = [Q]_\alpha a_+^\dagger a_+ = a_+^\dagger [Q]_\alpha a_+ \\
 & = [Q]_\alpha (-\delta + a_+ a_+^\dagger) - a_+^\dagger [Q]_\alpha a_+ \\
 \delta_\epsilon b_-^\dagger & = \int d^3p a_-^\dagger(p) = [Q]_\epsilon b_-^\dagger - b_-^\dagger [Q]_\epsilon \\
 [Q]_\epsilon b_- a_-^\dagger & = b_- [Q]_\epsilon b_-^\dagger - b_- b_-^\dagger [Q]_\epsilon \\
 [P]_\alpha b_- a_-^\dagger & = b_- [Q]_\alpha b_-^\dagger - (2\pi)^3 2E_p \delta^3 [Q]_\alpha \\
 & \quad - b_-^\dagger b_- [Q]
 \end{aligned}$$

when acting on  $\langle 0 | 10 \rangle$  the third term vanishes so we can ignore this factor.  $(I T I N K) \dots$

$$\int d^3p |a_+ b_+^\dagger - b_- a_-^\dagger| [P]_\alpha = 2 \int d^3p (2\pi)^3 2E_p \delta^3 [Q]_\alpha + a [Q] a^\dagger$$

last term acts as the 1-particle vacuum state  $\langle I | Q_\alpha | I \rangle$ , which I'm not sure if it should vanish... I'm assuming it must in this form.

Thus, overall I'm off by a factor of  $1/2$ .

Most likely not right, though.

$$\begin{aligned}
 [Q]_\alpha & = \int d^3p [P]_\alpha (a_+(p) b_-^\dagger(p) - b_-(p) a_+^\dagger(p)) \\
 [Q]_\alpha^\dagger & = \int d^3p [P]_\alpha^\dagger (a_-(p) b_+^\dagger(p) - b_+(p) a_-^\dagger(p))
 \end{aligned}$$

$$Q_m = \begin{pmatrix} [Q]_\alpha \\ [Q]_\alpha^\dagger \end{pmatrix}$$

Proof of correct form:

$$[\tilde{E}_m Q, Q] = \int d^3p [\epsilon(p) \alpha + \epsilon(p) \beta, Q]$$

(i)  $\delta_\epsilon a_-(q)$

$$\begin{aligned}
 & \int d^3p [\epsilon(p) \alpha + \epsilon(p) \beta, a_-(q)] \\
 & = a_+^\dagger b_+^\dagger a_- - a_+^\dagger a_+^\dagger b_+^\dagger \\
 & \quad - b_-^\dagger a_-^\dagger p a_- q + a_- q b_- p Q_-^\dagger p \quad (p=q) \\
 & = -b_+^\dagger p a_- q + a_+^\dagger p a_- q + (2\pi)^3 2E_p \delta^3 b_- p \\
 & = [\epsilon q] b_-(q)
 \end{aligned}$$

$\Rightarrow \alpha$ -term

no ~~0~~ - term:

$$[a - b^\dagger, -b + a^\dagger, a = 1q]$$

first vanishes as above

$$= -(b + a^\dagger) p a = q + a = q b + a +$$

Note for e

$[a_\pm, a_\pm^\dagger]$  Both must be + or -  
so this term vanishes since  $a - a^\dagger$  commute

\* Let us do  $\delta_e b_+$  for Good Measure:

$$\delta_e b_+ = \int d\vec{p} [\langle \epsilon p \rangle \alpha + \langle \epsilon p \rangle \beta, b_+(q)]$$

see  $\alpha$  is the only term which contains  $b_+^\dagger$  so this survives and yields

$$\langle \epsilon p \rangle a_+$$

$$\delta_e b_- : \beta \text{ contains } b_-^\dagger \text{ so}$$

$$\langle \epsilon p \rangle a_-$$

✓

$$\text{Ex 4.3} \{ |0\rangle_\alpha, \langle 0|_\alpha \}$$

$$= \int d\vec{p} d\vec{q} (|p\rangle_\alpha \langle q|_\alpha) \{ \alpha - \beta, \gamma - \delta \}$$

$$\alpha\gamma: \text{ all commute / anti commute} = 0$$

$$\beta\delta: = 0$$

thus, only cross terms survive:

$$= \{ \beta, \gamma \} = \{ \alpha, \delta \}$$

$$\{ \beta, \gamma \} = \{ b_{-p} a_{-p}^\dagger, a_{-q} b_{-q}^\dagger \}$$

$$= b_{-p} a_{-p}^\dagger a_{-q} b_{-q}^\dagger + a_{-q} b_{-q}^\dagger b_{-p} a_{-p}^\dagger$$

$$= b_{-p} (-\delta + a_{-q} a_{-p}^\dagger b_{-q}^\dagger) + \dots$$

$$= b_{-p} b_{-q}^\dagger \delta^3(p-q) + b_{-p} a_{-q} a_{-p}^\dagger b_{-q}^\dagger + \dots$$

$$= -b_{-p} b_{-q}^\dagger \delta^3(p-q) + a_{-q} (-b_{-q}^\dagger b_{-p} + \delta^3) a_{-p}^\dagger$$

$$+ \dots$$

$$= -(b_{-p} b_{-q}^\dagger - a_{-q} a_{-p}^\dagger) \delta^3(p-q)$$

$$\int d^3p \int d^3q \delta^3(\quad) = \int d^3p (b_{-p} b_{-p}^\dagger - a_{-p} a_{-p}^\dagger)$$

$$- \{a, \delta\} = \int d^3p (a_{+p} a_{+p}^\dagger - b_{+p} b_{+p}^\dagger)$$

$$\begin{aligned} \{[Q]_\alpha, [Q^\dagger]_\alpha\} &= (-P_{\alpha\beta}) \int d^3p \left[ (a_{+p} a_{+p}^\dagger - a_{-p} a_{-p}^\dagger) \right. \\ &\quad \left. - (b_{+p} b_{+p}^\dagger - b_{-p} b_{-p}^\dagger) \right] \\ &= (-P_{\alpha\beta}) \int d^3p \left[ (a_{+p}^\dagger a_{+p} - a_{-p}^\dagger a_{-p}) \right. \\ &\quad \left. + (b_{+p}^\dagger b_{+p} - b_{-p}^\dagger b_{-p}) \right] \end{aligned}$$

Number Operator for N is defined as

$$\left. \begin{aligned} N|n\rangle &= n|n\rangle \\ N|0\rangle &= 0 \\ N|1\rangle &= |1\rangle \quad \text{etc...} \end{aligned} \right\}$$

$$N_a^\pm = \int d^3p a_\pm^\dagger a_\pm$$

same for  $N_b^\pm$

$$\{[Q]_\alpha, [Q^\dagger]_\alpha\} = (-P_{\alpha\beta}) (N_a^+ - N_a^- + N_b^+ - N_b^-)$$

We find terms such as  $[Q, a_{-}(p)] = |p\rangle b_{-}(p)$   
and so the associated Bosonic & fermi 1-part.  
states are related i.e.

$$\begin{aligned} [Q, a_{-}(p)]^\dagger &= |p\rangle b_{-}^\dagger(p) |0\rangle \\ (a_{-}^\dagger Q^\dagger - Q^\dagger a_{-}^\dagger) |0\rangle &= |p\rangle (b_{-}^\dagger(p) |0\rangle) \\ a_{-}^\dagger ([Q^\dagger] |0\rangle) &= 0 \\ Q^\dagger (a_{-}^\dagger |0\rangle) &= -|p\rangle (b_{-}^\dagger(p) |0\rangle) \end{aligned}$$

∴ They are said to belong to the same

## Supermultiplet:

States in the same Supermult. must have the same mass.

CPT invariance requires us to have both  $\uparrow$  and  $\downarrow$  helicity sectors.

## Super Potential / Interactions:

$$\mathcal{L}_I = \frac{1}{2} g \phi \psi \psi + \frac{1}{2} g^* \bar{\phi} \psi^\dagger \psi^\dagger - \frac{1}{4} |g|^2 |\phi|^4$$

which modifies the susy transformations as

$$\delta \epsilon \phi = \epsilon \psi$$

$$\delta \epsilon \psi_a = -i \sigma_{a\dot{a}}^\mu \epsilon^{\dot{a}} \partial_\mu \phi + \frac{1}{2} g^* \bar{\phi}^2 \epsilon_a$$

$$\delta \epsilon \bar{\phi} = \epsilon^\dagger \psi^\dagger$$

$$\delta \epsilon \psi_a^\dagger = i \partial_\mu \bar{\phi} \epsilon^{\dot{b}} \sigma_{\dot{b}a}^\mu$$

Extended Susy: chiral Supermultiplet:  $\begin{matrix} \text{spin} & & \text{spin} \\ 0 & \oplus & 1/2 \end{matrix}$

$(\phi, \lambda, F)$   $F$ : off-shell scalar

and can repeat analysis for  $N > 1$  w susy particles of spin  $(S, s + 1/2)$

$N > 1$ :  $Q^A, Q_A^\dagger$   $A = 1, 2, \dots, N$

$$\{Q_a^A, Q_{B\dot{a}}^\dagger\} = -2 \sigma_{a\dot{a}}^\mu P_\mu \delta_B^A$$

• Can have a central charge  $Z^{AB}$   
 $\{Q_a^A, Q_b^B\} \sim Z^{AB} \epsilon_{ab}$

• Extended susy has an R-symmetry  $SU(N)$   
 $[T_R, Q] \neq 0$

Rotates  $Q^A$ 's into each other

$N=1$  examples:

|                     |                    |                           |         |
|---------------------|--------------------|---------------------------|---------|
| (i) Supermultiplets | $(0, 1/2)$         | photon                    | photino |
| (ii) Super QED      | $(\pm 1/2, \pm 1)$ | $(A_\mu, \lambda_\mu, ?)$ |         |

(iii) Super Yang Mills (SYM)  $(\pm 1/2, \pm 2) (\mathbb{B}_m, \psi_m, ?)$   
 gluon  $\rightarrow$   $\nwarrow$  gluino

\* Extended susy has  $2^N$  states in the massless supermult.

ex:  $N=2$  SYM  $S = -1 \quad -1/2 \quad 0$   
 • one supermult has: (1 gluon, 2 gluinos, scalar)  
 $\Rightarrow N=2$  has two bosonic states and two ferm. states

Max susy in  $D=4$  is  $N=4$ , to avoid spins greater than 1.

( $N=4$  SYM)

Amplitudes and susy Ward:  $N=1$

\*  $\mathcal{L} = \mathcal{L}_0 + \mathcal{L}_I$  already done for ex 2.8;  $\lambda = |g|^2$

$$A_4^{\text{tree}}(\phi\phi\bar{\phi}\bar{\phi}) = -|g|^2, \quad A_4^{\text{tree}}(\phi, f^-, f^+, \bar{\phi}) = -|g|^2 \langle 24 \rangle / \langle 34 \rangle$$

$$A_4^{\text{tree}}(f^- f^- f^+ f^+) = |g|^2 \frac{\langle 12 \rangle}{\langle 34 \rangle}$$

\* Note:

$$A_4^{\text{tree}}(\phi, f^-, f^+, \bar{\phi}) = \frac{\langle 24 \rangle}{\langle 34 \rangle} A_4^{\text{tree}}(\phi\phi\bar{\phi}\bar{\phi})$$

$$A_4^{\text{tree}}(f^- f^- f^+ f^+) = - \frac{\langle 12 \rangle}{\langle 24 \rangle} A_4^{\text{tree}}(\phi, f^-, f^+, \bar{\phi})$$

Susy ensures the above hold at all orders in the perturbation expansion. Lets see how this happens

$$\langle \mathcal{O}_1(p_1) \dots \mathcal{O}_n(p_n) | 0 \rangle \sim A_n$$

ex:

$$A_4(\phi, f^-, f^+, \bar{\phi}) = \langle \mathcal{O}_1(p_1) \mathcal{O}_2(p_2) \mathcal{O}_3^\dagger(p_3) \mathcal{O}_4^\dagger(p_4) | 0 \rangle$$

\* If the vacuum is supersymmetric  $Q|0\rangle = 0 = Q^\dagger|0\rangle$

Thus,

$$\langle \mathcal{O}_1 \dots \mathcal{O}_n | 0 \rangle = 0$$

$$\langle 0 | [Q^\dagger, \phi_1 \dots \phi_n] | 0 \rangle = \langle 0 | \phi_1 \dots \phi_n [Q^\dagger, 1] | 0 \rangle$$

$$= \langle 0 | ([Q^\dagger, \phi_1] + (-1)^{n_1} \phi_1 [Q^\dagger, \phi_2 \dots \phi_n]) | 0 \rangle$$

$$\left\{ \begin{array}{l} n_i = \begin{cases} 0: \phi_i: \text{Bosonic} \\ 1: \phi_i: \text{fermionic} \end{cases} \end{array} \right\}$$

$$= \langle 0 | [Q^\dagger, \phi_1] + (-1)^{n_1} \phi_1 ([Q^\dagger, \phi_2] + (-1)^{n_2} \phi_2 [Q^\dagger, \phi_3 \dots \phi_n]) \dots \phi_{n-1} ([Q^\dagger, \phi_n] + (-1)^{n_n} \phi_n [Q^\dagger, 1]) | 0 \rangle$$

$$0 = \sum_{i=1}^n (-1)^{n_i} \langle 0 | \phi_1(p_1) \dots [Q^\dagger, \phi_i(p_i)] \dots \phi_n(p_n) | 0 \rangle$$

and similarly for  $Q$

(Super symmetric Ward Identities)

• Valid at any order in pert. theory

example:

$$\langle 0 | [Q^\dagger, a_-(p_1) b_-(p_2) a_+(p_3) a_+(p_4)] | 0 \rangle$$

$$= \langle 0 | [Q^\dagger, a_-(p_1)] b_-(p_2) a_+^3 a_+^4 | 0 \rangle$$

$$+ \langle 0 | a_-^3 [Q^\dagger, b_-(p_2)] a_+^3 a_+^4 | 0 \rangle$$

$$= \langle 0 | a_-^3 b_-^2 [Q^\dagger, a_+^3] a_+^4 | 0 \rangle$$

$$= \langle 0 | a_-^3 b_-^2 a_+^3 [Q^\dagger, a_+^4] | 0 \rangle$$

where we found  $[Q^\dagger, a_-(p_1)] = 0$

$$[Q^\dagger, b_-(p_2)] = |p_2\rangle a_-(p_2)$$

$$[Q^\dagger, a_+^3] = |p_3\rangle b_+(p_3)$$

$$[Q^\dagger, a_+^4] = |p_4\rangle b_+(p_4)$$

$$= |2\rangle \langle 0 | a_-^2 a_+^3 a_+^4 | 0 \rangle - |3\rangle \langle 0 | a_-^2 b_-^2 a_+^3 a_+^4 | 0 \rangle$$

$$- |4\rangle \langle 0 | a_-^2 b_-^2 a_+^3 b_+^4 | 0 \rangle$$

$$0 = |2\rangle A_u(\phi\phi\bar{\phi}\bar{\phi}) - |3\rangle A_u(\phi, f^-, f^+, \bar{f})$$

$$- |4\rangle A_u(\phi, f^-, \bar{f}, \bar{f}^+)$$

( \* Each Ward id gives two relations: one from

Q† the other from Q

Obtain the two relations by projecting out states

$$(21): \langle 23 \rangle A_4(\phi, f^-, f^+, \bar{\phi}) = -\langle 24 \rangle A_4(\phi, f^-, \bar{\phi}, f^+)$$

$$A_4(\phi, f^-, \bar{\phi}, f^+) = \frac{-\langle 23 \rangle}{\langle 24 \rangle} A_4(\phi, f^-, f^+, \bar{\phi})$$

$$(41): \langle 24 \rangle A_4(\phi \phi \bar{\phi} \bar{\phi}) = \langle 34 \rangle A_4(\phi, f^-, f^+, \bar{\phi})$$

$$A_4(\phi, f^-, f^+, \bar{\phi}) = \frac{\langle 24 \rangle}{\langle 34 \rangle} A_4(\phi \phi \bar{\phi} \bar{\phi})$$

Can consider  $\langle 0 | [Q^\dagger, b_-^1 b_-^2 b_+^3, a_+^4] | 0 \rangle$

$$= \langle 11 \rangle \langle 0 | a_-^1 b_-^2 b_+^3 a_+^4 | 0 \rangle$$

$$+ \langle 12 \rangle \langle 0 | b_-^1 a_-^2 b_+^3 a_+^4 | 0 \rangle$$

$$+ \langle 14 \rangle \langle 0 | b_-^1 b_-^2 b_+^3 b_+^4 | 0 \rangle$$

$$(12): \langle 21 \rangle A_4(\bar{\phi} f^- f^+ \phi) = \langle 24 \rangle A_4(f^- f^- f^+ f^+)$$

$$A_4(f^- f^- f^+ f^+) = -\frac{\langle 12 \rangle}{\langle 24 \rangle} A_4(\bar{\phi} f^- f^+ \phi)$$

By crossing symm we should have

$$A_4(\phi f^- f^+ \bar{\phi}) = A_4(\bar{\phi} f^- f^+ \phi)$$

also since  $[a_+, a_-] = 0$  i.e. they commute

\* Amplitudes whose external states are related by susy are linked to each other through linear relationships called susy ward id.

$$\langle 0 | \mathcal{E}(Q, \dots, Q_n) | 0 \rangle = \delta_{\mathcal{E}} \langle \quad \rangle = 0$$

$$\text{ex: } \langle 0 | [Q^\dagger, a_-^1 a_-^2 b_+^3, a_+^4 a_+^5 a_+^6] | 0 \rangle$$

verbs

$$0 = \langle 13 \rangle A_6(\phi \phi \phi \bar{\phi} \bar{\phi} \bar{\phi}) - \langle 14 \rangle A_6(\phi \phi f^- f^+ \bar{\phi} \bar{\phi})$$

$\rightarrow KFS \rightarrow A_6(\phi\phi + \dots) = (FC) A_6(\phi\phi + \dots)$   
 • Web of linear relations: typical for "non-MHV" types

## $N=1$ susy: Gauge theory:

$N=1$  SYM:  $\mathcal{L} = -\frac{1}{4} \text{Tr}(F_{\mu\nu} F^{\mu\nu}) + i \text{Tr}(\lambda^\dagger \bar{\sigma}^\mu D_\mu \lambda)$

all fields are in the adjoint of  $SU(N)$

$$D_\mu \lambda = \partial_\mu \lambda - ig [A_\mu, \lambda]$$

$c_\pm$ : Gluon annihilation ops ( $\pm 1$  helicity)

$b_\pm$ : Gluino annh. ops ( $\pm 1/2$  helicity)

$$\begin{aligned} \text{Thus, } [\tilde{Q}, c_+ |i\rangle] &= 0 & [Q, c_+] &= [i] b_- |i\rangle \\ [\tilde{Q}, b_+ |i\rangle] &= |i\rangle c_+ |i\rangle & [Q, b_+] &= 0 \\ [\tilde{Q}, b_- |i\rangle] &= 0 & [Q, b_- |i\rangle] &= [i] c_- |i\rangle \\ [Q, c_- |i\rangle] &= |i\rangle b_- |i\rangle & [Q, c_-] &= 0 \end{aligned}$$

\* Now, since  $[\tilde{Q}, b_+] = 0$

$$\langle 0 | [\tilde{Q}, b_+ c_+ \dots c_{n+}] | 0 \rangle = |1\rangle \langle 0 | c_+ \dots c_{n+} | 0 \rangle$$

$$\dots \Delta_n [g^+ \dots g^+] = 0$$

SYM:  $\Delta_n^{L\text{-loop}} [g^+ g^+ \dots g^+] = \Delta_n^{L\text{-loop}} [g^- g^+ \dots g^+] = 0$

ex 4.8 For  $n=3$  an MHV:

$$\Delta_3 [1^- 2^+ 3^+] = \langle 12 \rangle^4 / \langle 12 \rangle \langle 23 \rangle \langle 31 \rangle$$

the Ward id:

$$\langle 0 | [Q^\dagger, b_1^1 c_2^2 c_3^3 c_4^4] | 0 \rangle = 0$$

(RED)

$$\begin{aligned}
&= \langle 0 | c_-^2 c_+^2 c_+^4 | 0 \rangle = \langle 0 | b_-^1 [Q^+, c_-^2] c_+^2 c_+^4 | 0 \rangle \\
&= A_3 [2^- 3^+ 4^+] = \langle 0 | b_-^1 (| 2 \rangle b_-^2 c_+^2 c_+^4 | 0 \rangle \\
&\Rightarrow A_3 [2^- 3^+ 4^+] = 0 \dots
\end{aligned}$$

ex 9.91: Claim: For  $N=1$  SYM

$$A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+] = \frac{\langle 12 \rangle}{\langle 13 \rangle} A_n[g^- g^- g^+ g^+ \dots g^+]$$

Proof

$$\begin{aligned}
&\langle 0 | [\tilde{Q}, c_-^1 c_-^2 b_+^3 c_+^4 \dots c_+^n] | 0 \rangle = 0 \\
&= | 1 \rangle \langle 0 | b_-^1 c_-^2 b_+^3 c_+^4 \dots c_+^n | 0 \rangle + | 2 \rangle \langle 0 | c_-^1 b_-^2 b_+^3 c_+^4 \dots c_+^n | 0 \rangle \\
&\quad + | 3 \rangle \langle 0 | c_-^1 c_-^2 c_+^3 c_+^4 \dots c_+^n | 0 \rangle = 0
\end{aligned}$$

$$(\langle 11 \rangle): \langle 12 \rangle A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+] = \langle 13 \rangle A_n[g^- g^- g^+ g^+ \dots g^+]$$

$\therefore$

$$* A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n[g^- g^- g^+ g^+ \dots g^+]$$

as claimed

Next relate  $A_n[\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+]$  to the MHV  
given  $A_n[g^- g^- g^+ g^+ \dots g^+]$

• How use the above to relate  $A[\lambda \lambda \lambda \lambda \dots]$   
to  $A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+]$

$$\langle 0 | [\tilde{Q}, c_-^1 b_-^2 b_+^3 b_+^4 \dots c_+^n] | 0 \rangle = 0$$

$$\begin{aligned}
&= | 1 \rangle A_n[\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] - | 3 \rangle A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+] \\
&\langle r 1 \rangle A_n[\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] = \langle r 3 \rangle A_n[g^- \lambda^- \lambda^+ g^+ \dots g^+]
\end{aligned}$$

$$\therefore A_n[\lambda^- \lambda^- \lambda^+ \lambda^+ g^+ \dots g^+] = \frac{\langle r 3 \rangle}{\langle r 1 \rangle} \frac{\langle 13 \rangle}{\langle 12 \rangle} A_n[g^- g^- g^+ g^+ \dots g^+]$$

where  $r = \{1, 2, \dots, n\} \setminus \{1, 3\}$

Q: Does  $N=1$  susy allow for relationships between

$$A_n[1^- 2^- 3^+ \dots n^+] \text{ and } A_n[1^- 2^+ 3^- \dots n^+]$$

$$A: \langle 0 | [\mathcal{Q}, c_-^1 c_-^2 c_+^3 \dots c_+^n] | 0 \rangle = 0$$

$$|1\rangle \langle 0 | b_-^1 c_-^2 c_+^3 c_+^4 \dots c_+^n | 0 \rangle + |2\rangle \langle 0 | c_-^1 b_-^2 c_+^3 \dots c_+^n | 0 \rangle = 0$$

$\Rightarrow$  Can't relate flipped helicity states

$\mathcal{Q}$ : MHV and NMHV?:

$$NMHV: A_n[1^- 2^- 3^- 4^+ \dots n^+]$$

$$\langle 0 | [\mathcal{Q}, c_-^1 c_-^2 c_-^3 c_+^4 \dots c_+^n]$$

$$|1\rangle \langle 0 | b_-^1 c_-^2 b_+^3 c_+^4 \dots c_+^n \rangle + |2\rangle \langle 0 | c_-^1 b_-^2 b_+^3 c_+^4 \dots c_+^n \rangle + |3\rangle \langle 0 | c_-^1 c_-^2 c_-^3 \dots c_+^n | 0 \rangle = 0$$

$$= |1\rangle A_n[1^- 2^- 3^- 4^+ \dots n^+] + |2\rangle A_n[1^- 2^- 3^+ 4^+ \dots n^+] + |3\rangle A_n[1^- 2^- 3^- 4^- \dots n^+]$$

No way to relate MHV to NMHV...

\* Adding susy to recursion relations requires:

- (i) Introduce on-shell superspace which groups the amplitudes into Super Amplitudes
- (ii) Incorporate a shift of a Grassmann super part of the on-shell superspace into the BCFW shift

$N=4$  SYM: on-shell superspace & SuperAmplitudes

$N=4$  Action:

$$S = \int d^4x \text{Tr} \left[ -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} - \frac{1}{2} (D_\mu \Phi_\pm)^2 + \frac{i}{2} \bar{\Psi} \not{D} \Psi + \frac{g}{2} \bar{\Psi} \Gamma^\mu [\Phi_\pm, \Psi] + (g^2/4) [\Phi_\pm, \Phi_\pm]^2 \right]$$

$$D_\mu \equiv \partial_\mu - ig[A_\mu, \cdot]$$

$\phi_I$ : six Real scalars  $I=1, \dots, 6$  of  $SO(6) \sim SU(4)$

R-symmetry:  $SU(4)$

\* All fields in the Adjoint of the general Gauge group  $SU(N)$

•  $[\phi_I, \phi_J]$ : Associated w matrix structure of  $SU(N)$

• Fermions encoded in 10-dim Maj-Weyl spinors  $\psi$   
 $\Rightarrow \Gamma^\mu$ : 10-dim Clifford alg.

Claim:  $N=4$  SYM follows from dim red. of  $N=1$  SYM  
 in 10d.

See: Schwarz "Supersymmetric Yang-Mills theories"

\* Group the six scalars into six complex fields i.e. elements of  $SU(4)$ :  $\phi^{AB} = -\phi^{BA}$ ,  $A, B = 1, \dots, 4$   
 which satisfy the self duality relation

$$\phi_{AB} = \frac{1}{2} \epsilon_{ABCD} \phi^{CD} \quad \phi \sim \mathbb{E} \text{ of } SU(4)$$

• Fermions: 10d fermion fields give  $4+4$  gluino states

$$\begin{array}{ccc} \lambda^A & \oplus & \bar{\lambda}_A \\ (u = +1/2) & & (u = -1/2) \\ \square & \oplus & \mathbb{E}^? \text{ of } SU(4) \end{array}$$

$\bar{\lambda}_A$ : anti-fundament rep of  $SU(4)$

\*  $V_0$  of the Scalar Potential at the vacuum is an order parameter of susy breaking:

$V_0 = 0$  : necessary for preserving susy  
 $V_0 > 0$  : breaks susy

\* In  $N=4$  SYM the scalar potential  $V = \frac{g^2}{4} [\phi_i, \phi_j]^2$   
 $\Rightarrow$  The theory has a moduli space of  $N=4$  supersymmetric vacua with  $[\phi_i, \phi_j] = 0$

• At the origin of the moduli space, where all the scalar vacuum expectation values  $\langle \phi^A \rangle = 0$ , all states are massless and the theory contains no dimensional parameters

{ Moduli-space: Space of states  $[\phi_i, \phi_j] = 0$  trivially, the  
• origin has  $\phi_i = 0 = \phi_j$   
Here,  $\hat{\phi}_i \sim \phi^A$  must be  $\langle 0 | \phi^A | 0 \rangle = 0$   
at the Quantum level. }

\* At  $\langle \phi^A \rangle = 0$  the theory is conformally invariant  
 $T_\mu^\mu = 0$  up to the trace-anomaly. In particular, the Beta fun vanishes at all orders in pert. theory, and there is no running coupling.

• Theory is invariant under the enlarged spacetime symmetry group,

Conformal Group:  $SO(2,4) \sim SU(2,2)$

Super Conformal Group:  $SU(2,2|4)$

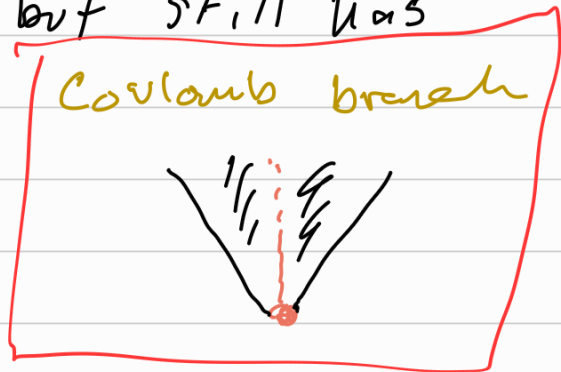
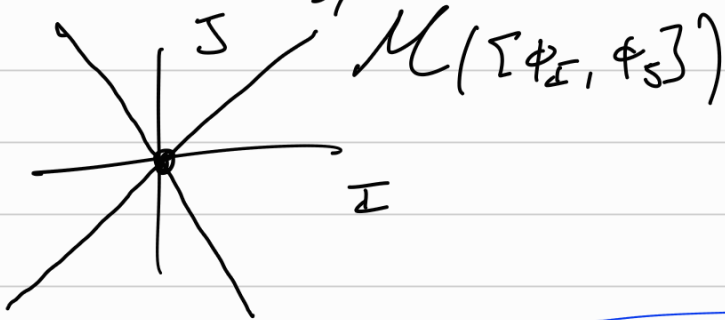
• Bosonic:  $SU(2,2)$  and R-sym group  $SU(4)$

• Fermionic 4-susy generators as well as their superconf. partners.

• All the scalar operators  $V(\phi)$  are conformally invariant

when the scalars acquire VEV's in such a way that full susy is preserved ( $[Q_I, \Phi_5] = 0$ ) the theory is said to be in the Coulomb branch  $\rightarrow$  i.e.  $\vec{I} \neq 0$

- The scale introduced by the scalar VEVs breaks the conf. sym, so the theory is NOT superconf. away from the origin but still has  $N=4$  susy.



\* Henceforth when we discuss amplitudes in  $N=4$  SYM this means the theory at the origin of the Moduli space

\* Given a Conf. theory we have no good notion of distance so the S-matrix is not, as usually known, well defined.

$\leadsto$  Consider a theory in  $(4-\epsilon)$ -dim

then the conf. symmetry is broken and the S-matrix is well defined.

- Inconv. for keeping on-shell symm manifest and thus better to look at the theory in the Coulomb branch and define  $N=4$  SYM

S-matrix as the zero-VEV limit of the Coulomb branch S-matrix

Spectrum and Susy Ward id.: (16 states)

|               |                |                       |                          |               |
|---------------|----------------|-----------------------|--------------------------|---------------|
| $\alpha$      | $q^A$          | $q^{AB}$              | $q^{ABC}$                | $q^{1234}$    |
| 1 gluon $g^+$ | 4 gluons $a^A$ | 6 scalars $\phi^{AB}$ | 4 gluons $\lambda^{ABC}$ | $-\lambda^D$  |
|               |                |                       |                          | 1 gluon $g^-$ |

| helicity: 1 | $1/2$     | 0         | $-1/2$    | -1        |
|-------------|-----------|-----------|-----------|-----------|
| 0           | $\square$ | $\square$ | $\square$ | $\square$ |
| $\pm 1$     | $\gamma$  | $\gamma$  | $\gamma$  | $\gamma$  |

\* Helicity  $h$ -states transform in the fully antisymmetric  $2(1-h)$  rep of  $SU(4)$

→ Multiplet is CPT self-conjugate

\* See text for Supercharges acting on cm.

ops

ex:  $[\bar{Q}_A, a(c)]_- = 0$

$$\{ |\bar{Q}_A, a(i)\rangle \} = |i\rangle \delta_A^B a(i)$$

$\tilde{Q}_A$  raises the helicity of ops by  $1/2$  and removes an index  $A$

ex 4.10 | From inspection we can obtain 4

d. Different Ward identities based on  $A = 1, \dots, 4$

\* Charge  $Q = 4$

$$g = : a^{1234},$$

$$\langle 0 | [\hat{Q}_4, g^- g^- \lambda^B g^+ \dots g^+] | 0 \rangle = 0$$

$$|0\rangle [\hat{Q}_n, a^{1234}] a^{1234} a_{B_1} \dots a_n |0\rangle + \dots$$

$$\{\widehat{Q}_4, a^{1234}|1\rangle\} = |1\rangle 4! \sum u^1 a^{234}|i\rangle$$

$$\begin{aligned} [\bar{Q}_n, a_n^\dagger] &= |3\rangle \delta_n^\beta a(3) \\ &= |3\rangle a(3) \end{aligned}$$

~~$$\therefore \langle 0 | a^{1234} (1) a^{1234} (2) a^4 (3) a (4) \dots a(n) | 0 \rangle = 0$$~~

$\rightarrow$   $1^2 \times 1 = 1^2 + 0$

$$(1) \rightarrow \Delta_n [g^- g^- g^- \dots g^-] + (2) \rightarrow \Delta_n [g^- g^- g^- \dots g^-] - (3) \rightarrow \Delta_n [g^- g^- g^- \dots g^-] = 0$$

Thus,

$$\Delta_n [g^-, \lambda^{123}, \lambda^4, g^+ \dots g^+] = \frac{\langle 13 \rangle}{\langle 12 \rangle} \Delta_n [g^- g^- g^+ \dots g^+]$$

$$\text{Claim: } \Delta_n [g^-, s^{12}, s^{34}, g^+ \dots g^+] = \frac{\langle 13 \rangle^2}{2 \langle 12 \rangle^2} \Delta_n [g^- g^- g^+ \dots g^+]$$

Proof: Use Ward identities:

$$[\bar{Q}_A, \lambda^{123}(z)] = |c\rangle 3! \delta_A^{[1} a^{23]}(z)$$

Thus, we want to keep  $\lambda^{123}$  as is and change  $\lambda^4(z) \rightarrow s^{34}(z)$

$$\langle 0 | [\bar{Q}_A, a^{1234} \lambda^{123} s^{34} a_u \dots a_n] | 0 \rangle = 0$$

$$\langle 0 | [\bar{Q}_A, a^{1234}] \lambda^{123} s^{34} a_u \dots a_n | 0 \rangle + \langle 0 | a^{1234} [\bar{Q}_A, \lambda^{123}] s^{34} a_u \dots a_n | 0 \rangle - \langle 0 | a^{1234} \lambda^{123} [\bar{Q}_A, s^{34}] a_u \dots a_n | 0 \rangle$$

$$(1) \rightarrow u! \langle 0 | \delta_A^{[1} a^{234]}(u) \lambda^{123} s^{34} a_u \dots a_n | 0 \rangle$$

$$+ (2) \rightarrow 3! \langle 0 | a^{1234} \delta_A^{[1} a^{23]}(2) s^{34}(3) a_u \dots a_n | 0 \rangle$$

$$= (3) \rightarrow \langle 0 | a^{1234} \lambda^{123}(2!) \delta_A^{[3} a^{4]}(3) a_u \dots a_n | 0 \rangle$$

\* To obtain the given relation we must have,  $A=3$

$$\therefore \delta_3^{[1} a^{234]}(3) = u! a^{134} / \delta_3^{[1} a^{23]}(2) = \frac{1}{3!} a^{12}(2)$$

$$\delta_3^{[3} a^{4]}(3) = \frac{1}{2!} a^4$$

$$(1) \rightarrow \langle 0 | a^{124} a^{123} s^{34} a_u \dots a_n | 0 \rangle$$

$$+ (2) \rightarrow \langle 0 | a^{1234} a^{12} a^{34} a_u \dots a_n | 0 \rangle$$

$$= (3) \rightarrow \langle 0 | a^{1234} a^{123} a^4 a_u \dots a_n | 0 \rangle = 0$$

$$\langle 12 \rangle \Delta_n [g^- s^{12} s^{34} g^+ \dots g^+] = \langle 13 \rangle \Delta_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+]$$

$$\Delta_n [g^- s^{12} s^{34} g^+ \dots g^+] = \frac{\langle 13 \rangle^2}{\langle 12 \rangle^2} \Delta_n [g^- g^- g^+ g^+ \dots g^+]$$

$$N=q \text{ SYM } \Delta_n [g^+ g^+ \dots g_i^- \dots g_i^- g^+ \dots g^+] = \frac{\langle ij \rangle^4}{\langle 12 \rangle^4} \Delta_n [g^- g^- g^+ g^+ \dots g^+]$$

Proof: Continue to use the above with  $g^-$  moved  $A=2$

$$\begin{aligned} \langle 0 | [\tilde{Q}, a^{1234} a^{12} a^{234} a_u \dots a_n] | 0 \rangle &= 0 \\ |1\rangle \langle 1| + \langle 0 | a^{1234} a^1 a^{234} a_u \dots a_n | 0 \rangle |2\rangle \\ &= |3\rangle \langle 0 | a^{1234} a^{12} a^{34} a_u \dots a_n | 0 \rangle = 0 \end{aligned}$$

$$\langle 12 | \Delta_n [g^- \lambda^1 \tilde{\lambda}^{234} g^+ \dots g^+] = \langle 13 | \Delta_n [g^- s^{12} s^{34} a_u \dots a_n]$$

$$\Delta_n [g^- \lambda^1 \tilde{\lambda}^{234} g^+ \dots g^+] = \frac{\langle 13 \rangle^3}{\langle 12 \rangle^3} \Delta_n [g^- g^- g^+ \dots g^+]$$

$$\begin{aligned} \langle 0 | [\tilde{Q}, a^{1234} a^1 a^{1234} a_u \dots a_n] | 0 \rangle &= 0 \quad A > 1 \\ \langle 0 | \lambda | 0 \rangle | 1 \rangle + | 2 \rangle \langle 0 | a^{1234} a_2 a^{1234} a_u \dots a_n | 0 \rangle \\ &= | 3 \rangle \langle 0 | a^{1234} a^1 a^{234} a_u \dots a_n | 0 \rangle \end{aligned}$$

$$\Delta_n [g^- g^+ g^- g^+ \dots g^+] = \frac{\langle 13 \rangle^4}{\langle 12 \rangle^4} \Delta_n [g^+ g^- g^+ \dots g^+]$$

Cycle,  $g^-$  w trace prop. to change  $g_i^+$

\* Note  $N=1$  is not CPT invariant, however  $N=4$  SYM is.

$N^k$  MHV classification:  $(k+2)$  negative helicity gluons

Def.  $N^k$  MHV Sector: all amplitudes in  $N=4$  SYM connected to the  $N^k$  MHV gluon amplitude via susy.

ex: all MHV amplitudes (of different particle species) are  $L$  MHV gluon amp.

→  $SU(4)_R$  ensures that the amp. vanishes unless the external states combine to the singlet ( $\underline{1}$ ) of  $SU(4)_R$

$N^k$  MHV: Requires  $(k+2)$  sets of  $\{1234\}$

→ Note for  $k=0$  this is exactly how the alternating MHV was determined

□ Each step contained the initial  $q^{1234}$  but also had  $\{1234\}$  split between the other two states  $P(2)$  and  $P(3)$ .

On-shell Superspace: Grassmann Variables  $\theta_a$ ,  $a=1, \dots, 4$

• which is an  $SU(4)$  label

\* We can define the  $N=4$  on-shell chiral Superfield  $\Omega$

which contains all 16 states

(8 Bosonic  $\oplus$  8 Fermionic)

$$\Omega = q^+ + \theta_a \lambda^a - \frac{1}{2!} \theta_a \theta_b S^{AB} - \frac{1}{3!} \theta_a \theta_b \theta_c \lambda^{ABC} + \theta_1 \theta_2 \theta_3 \theta_4 q^-$$

These signs are chosen s.t. the Grassmann differential ops

|       |           |       |                |                             |                                          |                                                       |
|-------|-----------|-------|----------------|-----------------------------|------------------------------------------|-------------------------------------------------------|
| $N=4$ | particle  | $q^+$ | $\lambda^A$    | $S^{AB}$                    | $\lambda^{ABC}$                          | $q^- = q^{1234}$                                      |
|       | operators | $1$   | $\partial_i^A$ | $\partial_i^A \partial_i^B$ | $\partial_i^A \partial_i^B \partial_i^C$ | $\partial_i^1 \partial_i^2 \partial_i^3 \partial_i^4$ |

Selects the associated state from  $\Omega(p_i)$

\* In the on-shell formalism, Supercharges are represented as

$$q_{\dot{A}a} \equiv [P]_{\dot{A}}^a \frac{\partial}{\partial \theta_a} \quad \Bigg| \quad q_{\dot{A}}^{\dot{a}} \equiv |p\rangle^{\dot{a}} \theta_a$$

\*  $Q_A$ : Supercharge in  $\mathbb{M}^4$  in superspace

For  $N=1$   $Q \sim |\bar{p}| p \rangle ( )$  so it's of no surprise that

$q$  is  $|p\rangle$  or  $|\bar{p}|$  Now also never said that

if would be  $\propto \eta_A$  since we now have  $SU(4)$  labels

\* These are on-shell

Claim:  $q$  and  $\tilde{q}$  satisfy the standard susy anticommutators

Proof:

$$\{q^{Aa}, \tilde{q}_B^{\dot{b}}\} = \left( [P]^a \frac{2}{m_A} \right) (|P\rangle^a \eta_B^{\dot{b}}) + \dots$$

$$= |P\rangle^{\dot{a}} [P]^a \delta_B^a = -\sigma_{\mu}^{\dot{a}a} P_{\mu} \delta_B^a \quad \checkmark$$

\* Note  $[\tilde{Q}, Q]$  shifts all terms and results in the commutation-relations of  $[\tilde{Q}, a(c)]$  etc....

ex: Relating  $N=1$  off-shell supercharges to  $N=4$  on-shell

\* For  $N=1$   $(x^{\mu}, \theta^a, \bar{\theta}^{\dot{a}})$  w/  $\{Q_a, Q_b\} = \{\bar{Q}_{\dot{a}}, \bar{Q}_{\dot{b}}\} = 0$   
 $\{Q_a, \bar{Q}_{\dot{a}}\} = i(\sigma^{\mu})_{a\dot{a}} P_{\mu}$

$$Q_a = \frac{\partial}{\partial \theta^a}, \quad \bar{Q}_{\dot{a}} = -\frac{\partial}{\partial \bar{\theta}^{\dot{a}}} + i\theta^a \sigma^{\mu}_{a\dot{a}} P_{\mu}$$

$$\{Q_a, \bar{Q}_{\dot{a}}\} = \left\{ \frac{\partial}{\partial \theta^a}, -\theta^b \sigma^{\mu}_{b\dot{a}} P_{\mu} \right\} = -(\sigma^{\mu})_{a\dot{a}} P_{\mu}$$

$\{P_a, P_{\dot{a}}\}$  does not contribute

off-shell:

$$\{P_a, \theta^b \sigma^{\mu}_{b\dot{a}} P_{\mu}\} = \sigma^{\mu}_{a\dot{a}} P_{\mu}$$

$\rightarrow$  If we go on-shell:  $P^2=0$ , then we know  $P_{a\dot{a}} = -|P\rangle [P]_{\dot{a}} = |P\rangle \langle P|_{\dot{a}}$

$$\{P_a, \theta^b |P\rangle_{\dot{b}} \langle P|_{\dot{a}}\} = |P\rangle_{\dot{a}} \langle P|_{\dot{a}}$$

\* We can introduce new Grassmann variables

$\Omega \equiv \theta^a [P]_a$  then  $\frac{\partial}{\partial \theta^a} = \frac{\partial \Omega}{\partial \theta^a} \frac{\partial}{\partial \Omega}$   
 or  $\frac{\partial}{\partial \theta^a} = [P]_a \frac{\partial}{\partial \Omega}$   $\textcircled{\ast}$   $= [P]_a \frac{\partial}{\partial \Omega}$

$\left\{ [P]_a \frac{\partial}{\partial \Omega}, \Omega \right\} = [P]_a$

and so by complex conj.

$\left\{ [P]^a \frac{\partial}{\partial \Omega}, [P]^a \Omega \right\} = [P]^a$

$\uparrow$   
 $N=1$  version of  $N=4$  of on-shell susy generators  $q^a, q^{\dot{a}}$ .

\* Note: If we dot ref. spinors  $|\bar{w}\rangle$  and  $|\omega\rangle$  s.t.  $\omega \neq p, \bar{w} \neq p$  we obtain

$\textcircled{\ast} \left\{ \frac{\partial}{\partial \Omega}, \Omega \right\} = 1$   $\textcircled{\ast}$

little group:  $\Omega = \theta^a [P]_a$ , under a little group rescaling  $\Omega \rightarrow t^{-1} \Omega$  since  $\theta^a$  is inert

Thus,

$\Omega \rightarrow t^{-2} \Omega$

$\lambda \sim t^{-1/2} \lambda$  since  $t^{-2u} \lambda \Rightarrow u = 1/2 \checkmark$   
 For  $N=4$  we have  $\theta^a \rightarrow \theta^a_a$  i.e. it gains an  $SU(4)$  label.

Superamplitudes: We can think of  $\Omega_i \equiv \Omega(p_i)$  as a "superwavefunction" for the  $i^{\text{th}}$  external particle of a color ordered superamplitude

$$\mathcal{A}_n[\Omega_1, \dots, \Omega_n]$$

Note from the restriction  $\mathcal{M}_A \mathcal{M}_A = 0$   
for  $A=1, \dots, 4$  we are restricted to having at most  $\mathcal{M}_A \mathcal{M}_B \mathcal{M}_C \mathcal{M}_D$

• Note  $\mathcal{M}$  also comes with a momenta index  $\mathcal{M}_{iA}$

•  $\mathcal{A}_n[\Omega_1, \dots, \Omega_n]$  will be a polynomial in  $\mathcal{M}_{iA}$  of varying degrees.

• We can project out desired Amplitudes from  $\mathcal{A}_n[\Omega]$

$$\mathcal{A}_n[1^+ \dots i^- \dots j^- \dots n^+] = \left( \prod_{A=1}^4 \frac{\partial}{\partial \mathcal{M}_{iA}} \right) \left( \prod_{B=1}^4 \frac{\partial}{\partial \mathcal{M}_{jB}} \right) \mathcal{A}_n[\Omega_1, \dots, \Omega_n] \Big|_{\mathcal{M}_{KC}=0}$$

Note

$\mathcal{M}_{i1} \mathcal{M}_{i2} \mathcal{M}_{i3} \mathcal{M}_{i4}$  term is  $S_i^-$  and if all other  $\mathcal{M}_{KC}=0$  then the only contribution from  $\Omega_n$  is  $S_n^+$

$$\mathcal{A}_n[S^{12} S^{34} 3^- 4^+ \dots n^+] = \left( \frac{\partial}{\partial \mathcal{M}_{11}} \frac{\partial}{\partial \mathcal{M}_{12}} \right) \left( \frac{\partial}{\partial \mathcal{M}_{23}} \right) \left( \frac{\partial}{\partial \mathcal{M}_{24}} \right) \times \left( \prod_{A=1}^4 \frac{\partial}{\partial \mathcal{M}_{3A}} \right) \mathcal{A}_n[\Omega_1, \dots, \Omega_n] \Big|_{\mathcal{M}_{KC}=0}$$

\* The order  $K$  of a Grassmann polynomial in  $\mathcal{M}$  corresponds to the  $N^K$ MHV Sector. Thus

$$\mathcal{A}_n = \mathcal{A}_n^{\text{MHV}} + \mathcal{A}_n^{\text{NMHV}} + \mathcal{A}_n^{\text{N}^2\text{MHV}} + \dots + \mathcal{A}_n^{\text{anti-MHV}}$$

$\chi_n^{MHV}$  : has Grassmann degree 8  
 $(n_{1i} \dots n_{ui})(n_{1j} \dots n_{uj})$

$$\chi_n^{NMHV} \sim (n_{1i} \dots n_{ui})(n_{1j} \dots n_{uj})(n_{1k} \dots n_{uk})$$

Degree 12

ex:

$$\chi_n^{MEV} = A_n [g^- g^- g^+ \dots g^+] (n_1)^4 (n_2)^4$$

$$+ A_n [g^- \lambda^{123} \lambda^u \dots g^+] (n_1)^4 (n_{21} n_{22} n_{23}) (n_{3u}) + \dots$$

• Second term is + since we obtain

$$\begin{aligned} \Omega_1 \Omega_2 \Omega_3 &= g^- n_1^u (-n_{21} n_{22} n_{23}) \lambda^{123} \\ &\quad \times n_{3u} \lambda^u \\ &= -g^- (-\lambda^{123} \lambda^u) (n_1^u n_{21} n_{22} n_{23} n_{3u}) \\ &= \oplus \quad \quad \quad \text{• Comes from anticom. of } \lambda^u \text{ and } n. \end{aligned}$$

\* Next, we look at the supersymmetric Ward identities in the language of on-shell superspace:

Supercharges:

$$Q^A \equiv \sum_{i=1}^n q_i^A = \sum_{i=1}^n [i | \frac{\partial}{\partial n_{iA}}$$

$$\tilde{Q}_A \equiv \sum_{i=1}^n q_A^\dagger = \sum_{i=1}^n |i\rangle n_{iA}$$

Ward identities:

$$Q^A \mathcal{L}_n = 0$$

$$\tilde{Q}_A \mathcal{L}_n = 0$$

ex 4.13

$$\begin{aligned} \mathcal{L}_n^{MHV} &= A_n [g^- g^- g^+ \dots g^+] (m_1)^4 (m_2)^4 + A_n [g^- \lambda^{123} \lambda^4 \dots g^+] (m_1)^4 (m_2, m_{22} m_{23}) m_{34} \\ &\neq A_n [\lambda^{123} g^- \lambda^4 g^+ \dots g^+] m_{11} m_{12} m_{13} (m_2)^4 m_{34} \\ &+ \dots i, j \text{ permutations} \end{aligned}$$

\* Ward states the coeff. of each independent Grassmann monomial in  $\tilde{Q}_A \mathcal{L}_n^{MHV}$  has to vanish.

pick  $A=4$ , then  $\tilde{Q}_4 = \sum_{i=1}^n |i\rangle m_{i4}$

• first term vanishes trivially for  $i=1,2$  however we still have  $\sum_{i=3}^n |i\rangle m_{i4}$

$$\begin{aligned} &= (m_1)^2 (m_2)^2 m_{34} [13\rangle A_n [g^- g^- g^+ \dots g^+] - A_n [g^- \lambda^{123} \lambda^4 g^+ \dots g^+] |2\rangle \\ &\quad - |1\rangle A_n [\lambda^{123} g^- \lambda^4 g^+ \dots g^+]] \\ &+ \sum_{i=4}^n |i\rangle m_{i4} \end{aligned}$$

Not sure how  $n>3$  terms in  $\sum_{i=4}^n$  vanishes....

the original component SUSY-Ward id. are manifest up to this extra term when we require

$$\tilde{Q}_A \mathcal{L}_n^{MHV} = 0$$

✓

\* The requirement

$$\{Q^A, \tilde{Q}_B\} \sim \sum_i p_i^\mu$$